

Solution Requirements

Date	9 july 2026
Team ID	
Project Name	StudyAI-Smart AI educate Companion&Quiz Generator
Maximum Marks	2 Marks

Define Functional and Non-Functional Requirements:

This document defines exactly what StudyAI will do -letting students upload study material and receive AI-generated summaries, flashcards, and quizzes- and how well it needs to perform, covering functional requirements like authentication and AI content generation, and non-functional requirements like response speed, security, and scalability.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users securely log in or sign up using Firebase Authentication with email/password, receiving a JWT token for session handling	High
2	Authorization levels	All authenticated users have equal access as students; each user can only view/edit their own uploaded content and scores	Medium
3	External interfaces	<i>Integrates with Groq API (Llama 3.3 70B Versatile) for AI content generation and Firebase Firestore for data storage</i>	High

4	Transactions processing	Handles file upload, text parsing, and AI generation requests as core transactions between frontend and backend	High
5	Reporting	Generates AI summaries, flashcards, quizzes, and analytics dashboards from uploaded study material	High
6	Business rules, etc.	Uploaded files must be PDF, DOCX, or TXT only; quiz includes MCQ, True/False, and Short Answer types	Medium
7	Compliance to laws or regulations	User credentials and data handled per Firebase's security standards; no sensitive data stored in plaintext	Medium
8	<i>Other</i>	Users can track quiz history and performance trends over time via the analytics dashboard	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	AI summary/flashcard/quiz generation should complete quickly using Groq's low-latency inference	Content generated in under 10 seconds
2	Scalability	Backend (Flask + Gunicorn) should handle multiple concurrent uploads and API requests	Supports up to 500 concurrent users
3	Security & Data Privacy	JWT-based auth, Flask-CORS protection, and environment-variable secrets for API keys	No plaintext secrets, secure token expiry
4	Reliability & Availability	Frontend on Vercel, backend on Render, both with high uptime	99% uptime per month
5	Usability & Accessibility	Simple, responsive UI built with React + Tailwind CSS	Mobile and desktop responsive
6	<i>Other</i>	Consistent structured AI output using well-engineered prompts	Maintainable prompt templates